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Comparative Time-Series Forecasting of Food Inflation using El Niño Indices: ARIMA vs. Prophet vs. LSTM

Guide: Prof. Vaishali Wadhe

Semester V | Department of AI & DS

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Introduction

Project context and key concepts

Food Inflation

Year-on-year percentage change in a food-price index, observed monthly.

Oceanic Niño Index (ONI)

ENSO-related sea-surface temperature anomaly indicator used as a climate input.

Forecasting Task

Compare models on a held-out monthly food-inflation test period.

Dataset used in the project

Period	Frequency	Observations	Core Variables	Final Test
2012–2025	Monthly	168	Food + ONI	24 months

The processed dataset contains no missing months or duplicate dates and covers January 2012 through December 2025.

Literature Review

Research / source	Contribution	Main finding or approach	Relevance to this project
Climate shocks in India [6]	Climate and food prices	Assesses drought, flood and other climate shocks using annual Indian data.	Motivates climate analysis; does not test our ONI forecasting benchmark.
Hochreiter & Schmidhuber [3]	Long Short-Term Memory	Introduces recurrent memory to learn sequential dependencies.	Foundation for both LSTM variants.
Taylor & Letham [4]	Prophet forecasting	Uses flexible trend and seasonality; supports external regressors.	Foundation for Prophet + ONI.
Hyndman & Koehler [5]	Forecast-error metrics	Examines accuracy metrics and the MASE measure.	Supports MAE, RMSE, sMAPE and MASE evaluation.
Statsmodels SARIMAX [7]	ARIMA with regressors	Models time-series dynamics with optional exogenous features.	Supports ARIMAX + lagged ONI.

Research Gap

Research gap: assess whether monthly ONI improves Indian food-inflation predictions beyond an inflation-only LSTM on a common 24-month test period. Data definitions: MoSPI [1] and NOAA [2].

Motivation

Why this forecasting comparison is useful within the project

The project is driven by one central question:

"Does adding the Oceanic Niño Index improve food-inflation forecasting beyond historical inflation patterns?"

Climate + Economic Signal

Combine food inflation history with ONI in the same forecasting study.

Model Comparison

Compare classical statistical, additive and deep-learning approaches.

Practical Delivery

Serve processed data, predictions and comparison metrics through FastAPI.

Problem Statement

Forecast monthly Indian food inflation using historical food-inflation observations and ONI, evaluate ARIMAX, Prophet, LSTM + ONI, and an inflation-only LSTM on January 2024–December 2025, and test whether ONI adds predictive value.

Input

Food inflation + ONI

Output

Held-out monthly predictions

Evaluation

MAE • RMSE • sMAPE • MASE

Objectives

01

Prepare Data

Derive a clean monthly food-inflation series and merge it with ONI.

02

Build Models

Implement ARIMAX, Prophet, LSTM + ONI and a univariate LSTM baseline.

03

Evaluate

Compare the selected models on the common January 2024–December 2025 test period.

04

Deliver Results

Compare MAE, RMSE, sMAPE and MASE; present stored results through FastAPI and a React dashboard.

Methodology

Processing, forecasting models and delivery architecture

01 Data sources
CPI/CFPI (Excel)
NOAA ONI (CSV)

→

02 Preprocess & merge
YoY fill; ONI cleanup
Monthly date alignment

→

03 Chronological split
Train: 2012–2023
Test: 2024–2025

→

04 Fit four models
ARIMAX, Prophet
Two LSTMs

↓ continued

05 Evaluate
MAE, RMSE
sMAPE, MASE

→

06 Save outputs
Predictions + metrics
Saved as CSV files

→

07 Display
FastAPI → React
Interactive dashboard

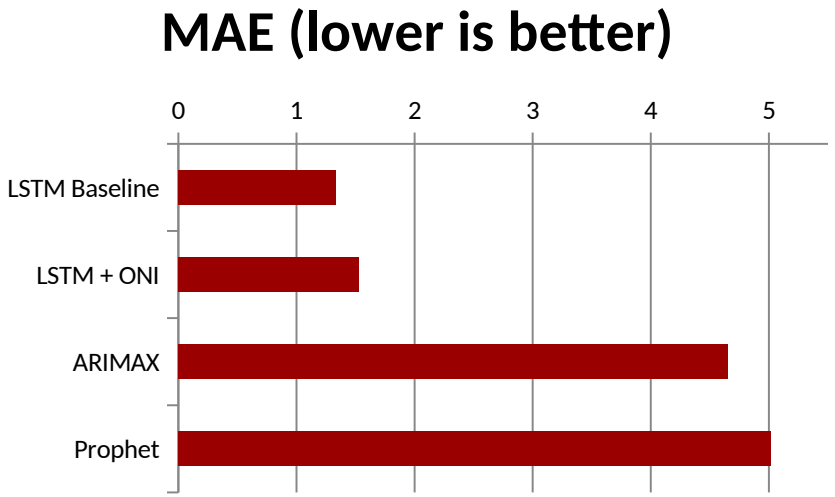
168 months (Jan 2012–Dec 2025) | 144 train / 24 test | ARIMAX(0,1,3) + lagged ONI; Prophet + ONI; LSTM + ONI (18-month lookback); baseline LSTM (12 months).

The API reads saved model outputs; the deployed website does not retrain models on each request.

Results and Discussion

Model comparison on the January 2024–December 2025 test period

Model	MAE	RMSE	sMAPE (%)	MASE
LSTM Baseline	**1.3335**	**1.6460**	**43.8900**	**0.9910**
LSTM + ONI	1.5245	1.9422	49.6371	1.1330
ARIMAX	4.6550	6.3055	85.4437	3.4594
Prophet	5.0169	6.6850	87.4038	3.7284



LSTM Baseline has the lowest error on all four metrics (MAE 1.3335); LSTM + ONI ranks second (MAE 1.5245). ONI did not improve LSTM accuracy here. Caveat: the models use different forecasting horizons and test-time information conditions.

Conclusion and Future Work

Conclusion

Processed 168 monthly inflation/ONI observations covering January 2012–December 2025.

Compared four models on the January 2024–December 2025 test window using MAE, RMSE, sMAPE and MASE.

LSTM Baseline ranked first (MAE 1.3335); LSTM + ONI ranked second (MAE 1.5245).

Implemented FastAPI endpoints and a React dashboard for displaying saved predictions and evaluation metrics.

Future Work

Extend the dataset with later CPI/CFPI and ONI observations and evaluate prospective forecasts.

Test rainfall, temperature, agricultural output and region-wise food-price indicators.

Standardize forecast horizons, input availability and rolling-origin validation across all models.

Investigate ensembles, tuning, uncertainty intervals, automated retraining and richer dashboard interactions.

References

Selected references (abbreviated for slides; full details in the project report)

[1] MoSPI, “Frequently Asked Questions: CPI and CFPI,” Government of India.

[2] NOAA PSL, “Climate Indices: Monthly Atmospheric and Ocean Time Series (ONI).”

[3] S. Hochreiter and J. Schmidhuber, “Long Short-Term Memory,” Neural Computation, 1997.

[4] S. J. Taylor and B. Letham, “Forecasting at Scale,” The American Statistician, 2018.

[5] R. J. Hyndman and A. B. Koehler, “Another look at measures of forecast accuracy,” IJF, 2006.

[6] M. Santhosh Kumar et al., “The effect of climate shocks on food price inflation, India,” J. Agric. Food Res., 2026.

[7] Statsmodels, “SARIMAX: Seasonal ARIMA with exogenous regressors,” API documentation.

Thank You

Questions & Answers

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